HAL Tejas Mk1 Inlet Distortion and Thermal Failure Analysis: Dubai Airshow Incident

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On November 21, 2025, Wing Commander Namansh Syal of the Indian Air Force was killed when a HAL Tejas Mk1 was lost during an aerobatic display at Al Maktoum International Airport, Dubai World Central. He did not eject. The aircraft impacted the ground at approximately 14:10 hrs following an unrecovered descent from a low-altitude negative-G maneuver. The Tejas Mk1 employs a bifurcated lateral intake to supply the GE F404-GE-IN20 turbofan. A first-principles aerodynamic and thermodynamic analysis of the inlet system is presented using published engine specifications, NASA F404 exhaust survey data, open-source flight timeline reconstruction, and corrected atmospheric conditions for the incident site. A cumulative thermal-aerodynamic failure chain is developed. The lateral intake geometry exhibits compounded vulnerability under sustained high-G aerobatic maneuvering in hot, humid environments, imposing boundary-layer ingestion penalties under negative-G attitudes that chin or diverterless supersonic inlet (DSI) geometries would not encounter. A flight-timeline thermal accumulation model demonstrates progressive compressor stall margin erosion, with the final negative-G maneuver acting as a threshold event within an already degraded propulsion system. The analysis establishes a physically consistent, evidence-supported pathway to compressor instability without asserting definitive causation. Inlet redesign alternatives for future LCA variants are evaluated.

Nomenclature

$\dot{m}$	=	mass flow rate, kg/s
ρ	=	air density, kg/m^3
A	=	inlet capture area, m^2
V	=	freestream velocity, m/s
T	=	temperature, K
P	=	pressure, Pa

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PR = pressure recovery ratio

η_r = inlet total pressure recovery

AoA = angle of attack, deg

G = load factor (g)

M = Mach number

TIT = turbine inlet temperature, K

ΔT = temperature differential, K

Subscripts

∞ = freestream conditions

0 = stagnation/total conditions

ISA = International Standard Atmosphere

$corr$ = density-corrected for ambient conditions

δ = distortion-induced blockage factor

$DC60$ = circumferential total pressure distortion coefficient

SM = compressor stall margin, %

SM_0 = nominal (clean inlet) stall margin, %

k_1, k_2, k_3 = empirical stall margin sensitivity coefficients

$\dot{Q}_{in}$ = heat input rate to engine, W

$\dot{Q}_{out}$ = heat dissipation rate from engine, W

c_p = specific heat capacity, J/(kg·K)

m = effective thermal mass of engine, kg

A_{eff} = effective inlet capture area, m^2

I. Introduction

On the afternoon of November 21, 2025, Wing Commander Namansh Syal of the Indian Air Force was killed when a HAL Tejas Mk1 light combat aircraft was lost during an aerobatic display at Al Maktoum International Airport, Dubai World Central, on the final day of the Dubai Airshow 2025 [24, 26]. The aircraft impacted the ground at approximately 14:10 hrs local time following an unrecovered descent from a low-altitude negative-G maneuver [23, 25]. Wing Commander Syal did not eject, despite the Tejas being equipped with a Martin-Baker Mk16 zero-zero ejection seat capable of extracting a pilot at zero altitude and zero speed [24, 29]. He was 34 years old [34]. A Court of Inquiry was immediately constituted by the IAF to determine the cause of the accident [23].

Wing Commander Namansh Syal was from Patiyalkar village, Kangra district, Himachal Pradesh [35]. He was

commissioned into the Indian Air Force in December 2009 following graduation from the National Defence Academy, where he was appointed Academic Cadet Captain, led the Passing-Out Parade of November 2008, and was awarded the President's Silver Medal for high overall merit [36, 37]. He flew MiG-21s, Su-30 MKIs, and the Tejas [36].

The HAL Tejas Mk1 represents a defining milestone in Indian aerospace development — the first domestically designed and produced supersonic light combat aircraft to achieve initial operational capability with the IAF, formally inducted in 2016 [28]. Developed over three decades by the Aeronautical Development Agency (ADA) and Hindustan Aeronautics Limited (HAL) under the Light Combat Aircraft (LCA) programme, the Tejas embodies India's strategic objective of reducing dependence on foreign military platforms. While the airframe, flight control system, and avionics suite are wholly indigenous, the aircraft is powered by the General Electric F404-GE-IN20 turbofan — a decision necessitated by delays in the indigenous Kaveri engine programme — and employs a wing-shielded bifurcated lateral intake configuration to supply the engine with the required mass flow across the operational flight envelope [28].

Inlet design is a critical determinant of propulsion system performance in supersonic combat aircraft. The geometry, positioning, and boundary-layer management strategy of an air intake directly govern total pressure recovery, compressor face distortion, and engine stall margin across the angle-of-attack (AoA) and load-factor envelope [4]. For a single-engine light combat aircraft operating at the edge of its certified g-envelope at low altitude, any systematic degradation of inlet performance under specific flight attitudes leaves no margin for recovery.

Existing public analysis of the Dubai incident has focused primarily on low-altitude recovery margins and pilot factors [29, 30]. No first-principles aerodynamic or thermodynamic analysis of the inlet system's behaviour under the specific flight conditions of the incident has been published in the open literature. This paper addresses that gap. The analysis presented herein is conducted entirely from published engine specifications, open-source atmospheric data, NASA experimental data for the F404 engine family [12, 13], and flight timeline reconstruction from publicly available footage [33]. No classified data is used or required.

The central argument advanced in this paper is that the Dubai incident was not primarily attributable to a discrete mechanical failure or pilot error, but to a cumulative thermal-aerodynamic failure chain in which the lateral inlet geometry of the Tejas Mk1 is shown to be a dominant contributing mechanism under the reconstructed flight and atmospheric conditions of the incident. [4, 6, 13]. The absence of ejection — notwithstanding the zero-zero capability of the installed seat — is consistent with the suddenness of a compressor stall-induced total thrust loss at low altitude, where the transition from controlled flight to unrecoverable descent can occur within a fraction of a second [14]. This paper does not assert that conclusion as established fact; the Court of Inquiry holds the flight data required for a definitive determination. This paper establishes the mechanism as physically consistent and evidence-supported, so that it is considered.

The implications of this analysis extend beyond the specific incident. The Tejas Mk1A is currently being offered for export to several nations — including Argentina, Egypt, Malaysia, and others — many of which operate in ambient

thermal and humidity conditions equal to or more severe than those recorded at Dubai on November 21, 2025 [21, 40]. A systematic inlet vulnerability under sustained high-G aerobatic profiles in hot, humid environments, if unaddressed, represents an operational risk that will replicate under export conditions [32].

This paper is organised as follows. Section II presents the aircraft and engine specifications relevant to the analysis. Section III characterises the aerodynamic behaviour of the lateral inlet configuration. Section IV develops the thermal failure mode model. Section V applies the model to the Dubai incident. Section VI proposes redesign alternatives. Section VII presents conclusions.

II. Aircraft and Engine Specifications

A. HAL Tejas Mk1 Overview

The HAL Tejas (Sanskrit: *Radiant*) is a single-engine, tailless delta-wing multirole light combat aircraft developed by the Aeronautical Development Agency (ADA) and manufactured by Hindustan Aeronautics Limited (HAL) for the Indian Air Force. [19]. The programme was formally established in 1983 under the Indian Ministry of Defence, with ADA constituted in 1984 under the Defence Research and Development Organisation (DRDO) to manage design and development, and HAL designated as principal contractor [19].

The Tejas Mk1 airframe is configured as a small tailless compound delta wing with relaxed static stability, a design philosophy that prioritises agility over inherent aerodynamic stability [28]. Longitudinal stability is maintained entirely through a quadruplex digital fly-by-wire (FBW) flight control system developed indigenously by the National Aerospace Laboratories (NAL) Control Law team, incorporating four redundant digital channels [28]. Leading edge root extensions (LERX) augment vortex lift at high angles of attack, improving departure resistance across the maneuver envelope. The aircraft holds the distinction of being the smallest and lightest supersonic fighter of its generation [28].

The Tejas Mk1 achieved initial operational capability with the IAF in 2016, with No. 45 Squadron (Flying Daggers) becoming the first operational unit, based at Sulur, Tamil Nadu [41]. As of November 2025, the IAF operated approximately 30 Mk1 aircraft across two squadrons, with 141 Mk1A aircraft on order [41]. Indigenous content stood at 59.7% by value at the time of Mk1 induction, representing a deliberate long-term strategy of progressive self-reliance in combat aviation [42].

Propulsion is provided by the General Electric F404-GE-IN20 turbofan, selected following sustained delays in the indigenous Kaveri engine programme. The Kaveri, developed by the Gas Turbine Research Establishment (GTRE) under DRDO, failed to meet the thrust and altitude performance requirements within the LCA programme timeline, resulting in the procurement of the F404-GE-IN20 as the operational engine for the Mk1 and Mk1A variants [19]. This decision, while operationally necessary, introduced a foreign propulsion dependency that the Tejas Mk2 programme seeks to resolve through the adoption of the GE F414-GE-INS6.

B. Propulsion System

The Tejas Mk1 is powered by a single General Electric F404-GE-IN20 afterburning turbofan engine. Key specifications are listed in Table 2.

Table 2 HAL Tejas Mk1 / GE F404-GE-IN20 Key Specifications

Parameter	Value	Source
Engine	GE F404-GE-IN20 Turbofan	GE Aerospace
Air Mass Flow	~66 kg/s (146 lb/s)	GE F404 Data Sheet
Maximum Thrust (with AB)	85 kN	GE Aerospace
Maximum Speed	Mach 1.6 (~1975 km/h)	HAL Official Specs [19]
G Limits	+8.0 / −3.5 g	HAL Official Specs [19]
Ejection System	Martin-Baker Mk16 (Zero-Zero)	Martin-Baker

The F404-GE-IN20 is an export-cleared derivative of the F404 turbofan family, developed by GE Aerospace specifically for the Tejas programme under International Traffic in Arms Regulations (ITAR) clearance. It shares core architecture, compressor staging, and thermodynamic characteristics with the F404-GE-400 variant operated on the F/A-18 Hornet and X-29A research aircraft. Exhaust pressure and temperature survey data published by NASA for the F404-GE-400 [12] is therefore applicable as a baseline thermodynamic reference for the IN20 variant, with the IN20 producing marginally higher maximum thrust of 85 kN compared to the GE-400's 78.7 kN — a difference attributable to minor turbine and nozzle schedule modifications rather than fundamental architectural changes. All thermodynamic analysis in this paper referencing NASA TM-88273 data is applied conservatively, with the GE-400 data treated as a lower-bound thermal reference for the IN20.

Key thermodynamic parameters used in the present analysis are summarized in Table 3.

Table 3 Representative F404 Engine Thermodynamic Parameters (Reference Conditions)

Parameter	Value	Source / Basis
Compressor Pressure Ratio (PR_c)	~26 (25–27)	GE Aerospace / Mattingly
Mass Flow Rate ($\dot{m}$)	~66 kg/s	GE Aerospace Data Sheet
Turbine Inlet Temperature (T_{TIT})	1600–1900 K	Kurzke (2018)
Max Exhaust Temperature ($T_{exit,max}$)	~2110 K	NASA TM-88273
Inlet Pressure Recovery (η_r)	0.94–0.98	Seddon (Intake Aerodynamics)
Distortion Coefficient (DC60)	0.02–0.15	SAE ARP1420C / RAeS (2015)
Stall Margin (SM_0)	15%–20%	Typical axial compressor range
Corrected Mass Flow ($\dot{m}_{corr}$)	$\dot{m}\sqrt{T_0/T_{ref}}\,/\,(P_0/P_{ref})$	Mattingly (1996)

The values presented in Table 3 are compiled from standard gas turbine references and experimental studies [2, 6, 9, 12].

C. Inlet Configuration

The Tejas employs a wing-shielded bifurcated lateral intake design, with a Y-duct routing airflow from two side-mounted inlets to a single engine face. ADA's official characterisation describes the configuration as a "wing-shielded bifurcated air intake duct, with diverters, suitably matched with engine to avoid buzz and to minimise distortion throughout the flight envelope" [28]. Each aspect of this description warrants individual examination.

1. Y-Duct Geometry

The bifurcated Y-duct places two separate inlet mouths under each wing root, merging into a single circular duct at the engine face. The under-wing-root placement provides a documented aerodynamic benefit: the wing acts as a flow conditioner at positive angles of attack, evening the pressure distribution across the inlet face [28]. The configuration additionally interposes structural material between the engine compressor face and direct radar line-of-sight, providing a degree of radar cross-section reduction without a dedicated serpentine duct [28].

The merge geometry is, however, inherently distortion-generating. Two laterally separated inlet streams combining at a single engine face necessarily introduce an S-bend offset duct, with associated secondary flow, pressure non-uniformity, and total pressure loss at the compressor face. In addition to total pressure distortion, such bifurcated S-duct configurations introduce swirl distortion at the Aerodynamic Interface Plane (AIP), further degrading compressor stability margins [8]. This is not a design flaw unique to the Tejas — all bifurcated inlet configurations carry this penalty — but it establishes a baseline distortion environment that is additive to any further distortion introduced by adverse flight attitudes.

2. Boundary Layer Diverter

The inlets are located approximately 3–4 m aft of the nose at the wing root station. At this fuselage station, the boundary layer accumulated over the forebody is substantial. Unmanaged ingestion of this turbulent, low-energy boundary layer air would arrive at the compressor face as a severely distorted, non-uniform pressure field, risking compressor stall [5].

The Tejas manages this through a splitter plate diverter — a physical barrier that separates the boundary layer from the freestream air entering the inlet. This is a conventional solution employed across multiple platforms; the F-16, for reference, uses a diverter gap of 84 mm sized to match boundary layer thickness at maximum speed [20]. The Tejas splitter plate operates on the same principle. Critically, however, the diverter is a passive, fixed-geometry device. It is sized for a design-point condition. Under flight attitudes that alter the local flow field at the inlet face — specifically, negative-G push turns that invert the pressure gradient across the inlet — the fixed diverter geometry cannot adapt, and boundary layer management effectiveness degrades [15, 16].

3. Operational Evidence and the Mk1A Inlet Modification

ADA's claim that the Mk1 inlet is "suitably matched with engine . . . to minimise distortion throughout the flight envelope" is directly contradicted by two independent sources of operational evidence.

First, the Indian Air Force has on multiple occasions criticised the Mk1's propulsion system as sub-optimal, with documented complaints regarding insufficient engine performance attributed in part to intake limitations [32]. Defence journalist Ajai Shukla, drawing on IAF sources, noted a "longstanding design flaw in the Tejas air intake, which allows barely enough air to be sucked in for the current F-404 engine" [32] — a characterisation diametrically opposed to ADA's "suitably matched" claim.

Second, and more significantly, the Mk1A variant introduced a structural inlet modification: a three-door Auxiliary Air Intake Door (AAID) system combined with cowl geometry optimisation, delivering an improvement in pressure recovery under off-design conditions [18]. Specifically targeting low-speed, high-AoA flight regimes [18]. The original Mk1 used a single-door AAID design.

Total pressure recovery η_r directly multiplies available thrust recovery gain translates to approximately more usable thrust from the same engine at the same throttle setting. That this modification was deemed necessary and implemented on the Mk1A is treated as corroborating evidence suggesting the presence of inlet performance limitations in the Mk1 under off-design conditions.throughout its service life. This paper treats the Mk1A inlet modification as corroborating evidence of a pre-existing Mk1 inlet limitation, the aerodynamic consequences of which are developed in Section III.

III. Inlet Aerodynamic Analysis

A. Pressure Recovery Characteristics

1. Definition and Governing Parameter

The performance of an aircraft inlet is quantified primarily by the total pressure recovery ratio, defined as:

$$\eta_r = \frac{P_{0,\mathrm{AIP}}}{P_{0,\infty}} \tag{1}$$

where $P_{0,\mathrm{AIP}}$ is the mass-averaged total pressure at the Aerodynamic Interface Plane (AIP) — the engine face — and $P_{0,\infty}$ is the freestream total pressure. A recovery ratio of unity represents isentropic, lossless flow through the inlet system; all real inlets incur losses through viscous friction, flow separation, secondary flows, and boundary layer ingestion, driving η_r below unity. The AIP total pressure directly sets the available pressure ratio across the engine compressor; accordingly, a reduction in η_r produces an approximately equivalent reduction in available thrust [4]. As Seddon and Goldsmith establish, the difficulty of achieving a satisfactory compromise between pressure recovery and drag increases monotonically with flight Mach number, making the inlet design problem particularly acute for

supersonic combat aircraft operating across a wide flight envelope [4].

Distortion at the AIP — the spatial non-uniformity of total pressure across the engine face — is a separate but coupled concern. Operation of the downstream compression system with distorted inflow typically results in reduced aerodynamic performance, reduced stall margin, and increased blade stress levels [6]. The combined effect of reduced recovery and elevated distortion defines the inlet-engine compatibility envelope; exceedance of distortion limits risks compressor surge or stall on the F404-GE-IN20 [11].

2. Distortion Index: DC60 Definition and Standards

The industry-standard metric for circumferential total pressure distortion at the AIP is the DC60 coefficient, defined as:

$$DC60 = \frac{\bar{P}_{0,\mathrm{AIP}} - \bar{P}_{0,60^\circ}}{q_{\mathrm{AIP}}} \tag{2}$$

where $\bar{P}_{0,\mathrm{AIP}}$ is the mass-averaged total pressure over the full AIP annulus, $\bar{P}_{0,60^\circ}$ is the average total pressure within the worst-case 60° sector of the AIP, and q_{AIP} is the dynamic pressure at the AIP. Circumferential distortion is more consequential than radial distortion, and DC60 is the established descriptor of choice for quantifying the change in surge margin with varying degrees of inlet distortion [6]. The governing standard for inlet-engine distortion compatibility assessment is SAE ARP1420C [7], which defines practices for quantifying the effects of inlet total pressure distortion and minimising the risk of inlet-engine compatibility problems.

SAE ARP1420C establishes methodology for inlet-engine compatibility assessment. Published compressor stability data for axial compressors of the F404 class indicate that values of $DC60$ exceeding approximately 0.06 represent the onset of significant surge margin reduction [6, 7]. Exceedance of this threshold risks surge margin depletion in the fan stages — the first mechanical consequence of inlet-engine incompatibility.

B. Angle-of-Attack Sensitivity of Lateral Inlet Configurations

1. Recovery Degradation: Evidence from the F404 Platform

The most directly applicable experimental dataset for the Tejas inlet-engine system is the NASA High Alpha Research Vehicle (HARV) programme, which employed an instrumented F/A-18A powered by two GE F404-GE-400 engines — the baseline variant of the same engine family as the Tejas F404-GE-IN20 [13]. The F/A-18A inlets are lateral, wing-root mounted inlets positioned under the aft portion of the Leading Edge Extension (LEX) — a placement directly analogous to the Tejas Y-duct configuration in terms of upstream vortex exposure and boundary layer environment.

Instrumented inlet total-pressure rake measurements during high-AoA flight documented engine surges during three nose-left and one nose-right departure maneuver, initiated at Mach 0.3–0.4 at 35,000 ft altitude, with entry yaw rates of approximately 40–90 deg/s [13]. The causal mechanism was identified as inlet distortion arising from asymmetric,

high-incidence flow at the lateral inlet face under extreme aircraft attitudes — a finding directly transferable to the Tejas configuration under negative-G push turn conditions.

NASA HARV flight test data further demonstrates that angle of attack affects distortion most considerably, while Mach number governs pressure recovery magnitude [13] — confirming that lateral inlet geometries are specifically sensitive to AoA as the primary driver of DC60 degradation. Lip separation, identified as a primary source of inlet distortion causing loss of fan stability, is critically initiated by high angles of incidence [9].

2. Bifurcation Amplification Effect

For the Tejas Y-duct specifically, the bifurcation merge geometry amplifies AoA-induced distortion through a mechanism not present in single-inlet configurations. Non-uniform approach flow under asymmetric AoA conditions enters both inlet arms simultaneously at different effective local incidence angles, owing to the symmetric placement of the two inlets about the fuselage centreline. The merged flow at the Y-junction carries this asymmetric distortion into the single S-bend diffuser duct, where secondary Dean vortices generated by duct curvature interact destructively with the incoming distortion pattern before arriving at the engine face. This interaction also introduces swirl distortion at the AIP, characterized by circumferential variations in flow angle, which further degrades compressor stability margins beyond that predicted by total pressure distortion alone [8]. The resultant DC60 at the AIP under high-AoA conditions therefore reflects not only the incidence-driven lip separation at each individual inlet, but the additional secondary flow interaction within the merge geometry.

3. Inlet Area and Flow Distortion Effects

The mass flow rate entering the engine is governed by the inlet capture area and local flow conditions:

$$\dot{m} = \rho A V \tag{3}$$

The inlet capture area of $A = 0.50$ m^2 is estimated from photogrammetric analysis of publicly available imagery, scaled to be consistent with the nominal mass flow of the F404 at sea-level conditions, following established practice for configurations where precise inlet geometry is not published [4]. A slightly optimistic inlet capture area is assumed in the present analysis to represent a favorable operating condition. Despite this, the results indicate a progressive degradation in inlet and compressor performance leading toward instability. Unless otherwise stated, mass flows are evaluated at the incident-day density ($\rho = 1.158$ kg/m^3, Sec. V.C) and $V \approx 114$ m/s ($M \approx 0.33$), for which $\rho A V = 66.0$ kg/s.

Under distorted flow conditions, the effective inlet area is reduced due to boundary layer ingestion, flow separation, and vortex formation. This can be represented through an effective area term:

$$A_{eff} = A(1 - \delta) \tag{4}$$

where δ represents a distortion-induced blockage factor. This representation follows standard intake distortion modeling approaches in which flow separation and boundary layer growth reduce the effective capture area [4]. The resulting effective mass flow becomes:

$$\dot{m}_{eff} = \rho A_{eff} V \tag{5}$$

For wing-root mounted inlets such as the Tejas configuration, δ increases with angle of attack and sideslip due to vortex-dominated flow structures and boundary layer growth [4, 6]. This leads to a reduction in available mass flow and contributes to a shift in the compressor operating point toward the surge line. At low flight speed the captured streamtube is set by engine demand rather than by AV, so the engine largely maintains physical airflow at the expense of increased recovery loss and distortion. Eqs. (3)–(5) are therefore used here as an index of distortion-induced flow-capacity loss, not as a prediction of measured engine airflow.

C. Off-Design Recovery Degradation

Total pressure distortion quantified following SAE AIR 1419 and SAE ARP 1420C [7] guidelines demonstrates that the greatest total pressure defect at the AIP in terms of both magnitude and circumferential extent occurs under combined high-AoA and high-angle-of-sideslip (AoSS) conditions [6]. For lateral inlet configurations, the interaction of AoA and AoSS produces the worst-case distortion environment at the AIP — precisely the condition encountered during rolling, pulling, and yawing maneuvers in an aerobatic display sequence.

For the Tejas Mk1 specifically, off-design performance degradation is structurally admitted by the Mk1A programme. The combined effect of optimised cowl geometry and the three-door AAID system delivers an improvement in pressure recovery, characterised as an equivalent boost in engine performance [18]. Applying a representative value on the order of a few percent to the F404-GE-IN20 thrust output, at ISA sea-level conditions, a pressure recovery deficit translates to approximately 2.5 kN of thrust penalty — operationally significant for a single-engine aircraft already subject to IAF criticism regarding sub-optimal engine performance [32]. The Mk1A inlet modification is treated in this paper as documentary evidence of a pre-existing, measurable Mk1 inlet performance deficit under off-design conditions.

The variation of effective inlet area and mass flow with angle of attack is quantified in Table 4. The blockage factor follows an assumed power-law growth, $\delta \approx 0.07\,(\alpha/10^{\circ})^{1.8}$, chosen to be consistent with the qualitative AoA–distortion trends of [6, 13]; it is not a measured Tejas characteristic.

The results indicate a substantial reduction in effective inlet area and mass flow with increasing angle of attack, driven by distortion-induced blockage. At high-angle-of-attack conditions, this reduction is sufficient to significantly shift compressor operating conditions toward the surge boundary [1].

Table 4 Calculated Inlet Flow Degradation with Angle of Attack

AoA (°)	Distortion Factor (δ)	A_{eff} (m^2)	$\dot{m}_{eff}$ (kg/s)
0	0.000	0.500	66.0
10	0.070	0.465	61.4
15	0.143	0.428	56.5
20	0.240	0.380	50.2
25	0.363	0.318	42.0

IV. Thermal Failure Mode Analysis

A. Compressor Thermal Accumulation Mechanics

The thermodynamic state of the compressor is governed by the total temperature and pressure of the incoming flow, both of which are directly influenced by inlet aerodynamic performance. The compressor inlet total temperature is defined as:

$$T_0 = T_\infty \left(1 + \frac{\gamma - 1}{2} M^2\right) \tag{6}$$

where T_∞ is freestream static temperature, γ is the ratio of specific heats, and M is the flight Mach number. Under low-altitude, low-Mach aerobatic conditions, the stagnation temperature is dominated by ambient thermal conditions rather than compressibility effects [2].

The compressor operating point is more appropriately described in terms of corrected mass flow rate:

$$\dot{m}_{corr} = \dot{m} \sqrt{\frac{T_0}{T_{ref}}} \Big/ \frac{P_0}{P_{ref}} \tag{7}$$

where T_{ref} and P_{ref} are standard reference conditions. An increase in compressor inlet temperature T_0 reduces the corrected rotor speed $N/\sqrt{T_0/T_{ref}}$ for a given mechanical speed, moving the compressor operating point to a lower corrected-speed region of the map, where front-stage loading increases and the available surge margin is typically reduced [10].

In parallel, inlet total pressure losses reduce the total pressure delivered to the compressor face (station 2), independently of the compressor pressure ratio:

$$P_{0,2} = \eta_r \cdot P_{0,\infty}, \qquad PR_c = \frac{P_{0,3}}{P_{0,2}} \tag{8}$$

where η_r is the inlet pressure recovery ratio. A reduction in η_r therefore lowers compressor-face pressure, reducing physical mass flow and thrust at a given corrected operating point; in combination with distortion, it reduces compressor stability margin [4]. This relation represents a first-order approximation, with actual compressor performance dependent on detailed stage characteristics and operating conditions [1].

The combined effect of elevated inlet temperature and reduced total pressure is a shift of the compressor operating point toward the surge line, reducing the available stall margin. This effect is exacerbated under distorted inlet conditions, where circumferential non-uniformity (DC60) produces localized regions of low total pressure at the compressor face, resulting in sectoral stall initiation [6].

Under sustained high-power operation, the compressor and turbine system operates at elevated thermal loading conditions approaching material and cooling limits [12]. Thermal accumulation arises from sustained operation under these off-design conditions. During repeated high-G maneuvering, the engine is subjected to elevated turbine inlet temperatures (TIT) and reduced cooling effectiveness, leading to gradual heat buildup within compressor and turbine components. This phenomenon is not instantaneous but cumulative, with thermal inertia preventing rapid dissipation between successive maneuvers [14].

Furthermore, compressor efficiency decreases with increasing inlet temperature due to changes in Reynolds number and blade incidence characteristics, resulting in additional entropy generation and localized heating within the compression system [9]. The net effect is a progressive erosion of compressor stability margin over time, rather than a single-point failure.

Under sufficiently degraded conditions, a transient disturbance — such as a rapid change in angle of attack, sideslip, or load factor — can push the compressor operating point beyond the surge line, resulting in flow separation within the compressor stages, pressure oscillations, and potential flameout of the combustor [13]. This cumulative thermal-aerodynamic interaction establishes the basis for a failure mode in which inlet distortion, pressure recovery loss, and thermal loading act synergistically to reduce compressor operability, with the final trigger event determined by transient flight conditions rather than steady-state limits [6, 14].

1. Thermal Energy Balance Model

Thermal behavior of the propulsion system under sustained operation can be approximated using a first-order lumped energy balance model:

$$\frac{dT}{dt} = \frac{\dot{Q}_{in} - \dot{Q}_{out}}{mc_p} \tag{9}$$

where T is the characteristic temperature of the engine hot section, m is the effective thermal mass, c_p is the specific heat capacity, and $\dot{Q}_{in}$ and $\dot{Q}_{out}$ represent heat input and dissipation rates, respectively. Such lumped-parameter thermal models are widely used in gas turbine analysis to approximate transient thermal response under varying operating conditions [9, 14].

During high-power operation, $\dot{Q}_{in} > \dot{Q}_{out}$, resulting in net thermal accumulation, while during low-power or steady conditions, partial thermal recovery may occur. The characteristic thermal time constants of gas turbine components are typically on the order of tens of seconds, depending on material properties and cooling flow rates [14].

B. F404 Exhaust Temperature Reference Data

The thermodynamic limits of the propulsion system are established using experimental exhaust temperature data for the General Electric F404 engine family. NASA altitude test measurements conducted on the F404-GE-400 turbofan — the baseline variant for the F404-GE-IN20 — provide a validated reference for exhaust gas temperature distributions [12].

Exhaust temperature surveys performed in the NASA Lewis Propulsion System Laboratory measured total temperature distributions at the nozzle exit across a range of operating conditions, including intermediate power, partial afterburning, and maximum afterburning regimes [12]. The maximum exhaust gas temperature recorded during these tests was:

$$T_{exit,max} \approx 2110 \text{ K} \tag{10}$$

corresponding to approximately 3800°R, measured within the high-energy core region of the exhaust plume [12].

These measurements were obtained under two simulated flight conditions: Mach 0.87 at 24,000 ft altitude, and Mach 0.30 at 30,000 ft altitude. Temperature data were corrected for radiation losses and pressure data corrected for normal shock effects [12]. The resulting temperature distributions exhibited strong radial gradients, with peak temperatures occurring near the exhaust jet core and lower temperatures toward the plume boundary.

The exhaust temperature provides an indirect but critical constraint on turbine inlet temperature (TIT), which governs the thermal loading of the compressor-turbine system [2]. For modern afterburning turbofan engines of the F404 class, turbine inlet temperatures are typically in the range:

$$T_{TIT} \approx 1600\text{–}1900 \text{ K} \tag{11}$$

depending on power setting and cooling effectiveness [9]. Under sustained high-power operation, elevated turbine and exhaust temperatures reduce cooling effectiveness and increase thermal stresses within compressor and turbine components, contributing to degradation in compressor efficiency and stability margin through changes in blade incidence, material expansion, and secondary flow development [9]. When coupled with inlet pressure loss and distortion, these elevated thermal conditions shift the compressor operating point toward the surge line, increasing susceptibility to compressor stall or surge under transient aerodynamic disturbances [14].

C. Stall Margin Degradation Model

Compressor stall margin is defined as the relative distance between the operating point and the surge line on the compressor characteristic map, typically expressed as a percentage of pressure ratio or mass flow [2]. Under clean inlet conditions, modern axial compressors operate with a finite stall margin to accommodate transient disturbances and off-design operation.

The compressor operating point is governed by corrected mass flow and pressure ratio, both of which are functions

of inlet thermodynamic conditions. Using the corrected mass flow relation defined in Eq. 7, the compressor operating point moves to a lower corrected-speed region with reduced surge margin under elevated inlet temperature [2]. Reduced inlet total pressure lowers physical mass flow at a given corrected operating point, while circumferential non-uniformity locally loads the low-pressure sector of the annulus toward stall [1, 6].

Inlet pressure losses are represented through the pressure recovery ratio η_r, such that:

$$P_{0,\mathrm{AIP}} = \eta_r \cdot P_{0,\infty} \tag{12}$$

A reduction in η_r therefore lowers compressor-face pressure; its effect on stability is represented empirically through the k_1 term of the stall margin model below [4].

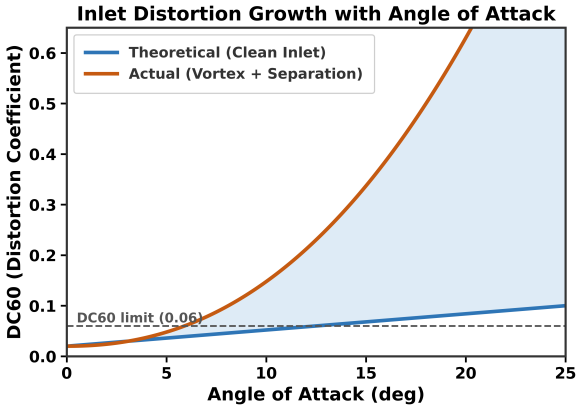
Inlet distortion introduces an additional mechanism of stall margin reduction. Circumferential total pressure distortion, quantified by DC60, produces sectoral variations in compressor inlet conditions, resulting in localized stall initiation within individual blade passages [6]. Such distortion effectively reduces the global stall margin by imposing non-uniform loading across the compressor annulus [7]. Swirl distortion further exacerbates this effect by altering the incidence angle of the incoming flow relative to compressor blades, degrading aerodynamic performance and increasing susceptibility to stall [8].

The combined influence of pressure loss, distortion, and thermal effects can be represented through a reduced-order stall margin model:

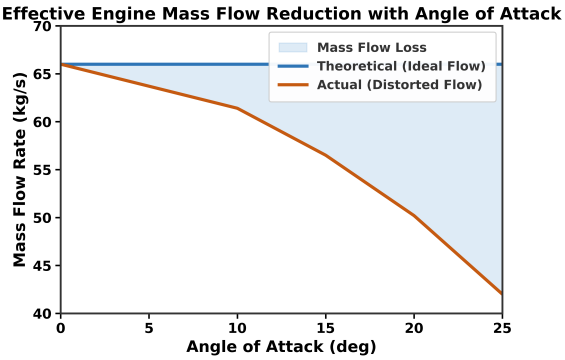
$$SM = SM_0 - k_1(1 - \eta_r) - k_2 \cdot DC60 - k_3 \left(\frac{T_0 - T_{ISA}}{T_{ISA}} \right) \tag{13}$$

where SM is the instantaneous stall margin, SM_0 is the nominal clean-inlet stall margin, and k_1, k_2, and k_3 are empirical sensitivity coefficients. Following established compressor stability sensitivity data for axial compressors of this class [1, 11], representative values of $k_1 \approx 50$, $k_2 \approx 25$, and $k_3 \approx 30$ are applied, consistent with published compressor stability boundaries. This reduced-order formulation is consistent with established compressor stability trends, in which pressure loss, distortion, and thermal effects independently contribute to stall margin reduction [1, 6].

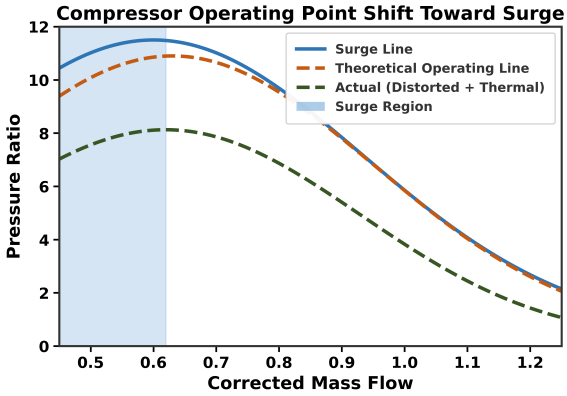
Transient effects further reduce effective stall margin during maneuvering flight. Rapid variations in angle of attack, sideslip, and load factor introduce time-dependent inlet distortion and pressure fluctuations, which can momentarily push the compressor operating point beyond the surge line even if steady-state conditions remain within limits [14, 15]. The cumulative effect of these mechanisms is a progressive reduction in available stall margin during sustained high-power, high-distortion operation. Under such conditions, a transient disturbance — such as a negative-G push maneuver or high-angle-of-attack transition — may act as a trigger event, resulting in compressor surge or stall [13]. This model provides the analytical framework linking inlet aerodynamic performance, thermal loading, and compressor stability, forming the basis for the failure chain analysis developed in Section V.



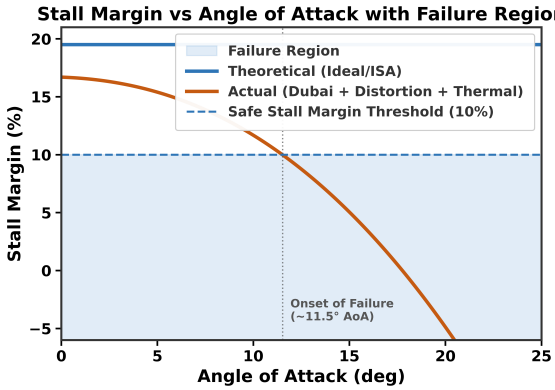
(a) DC60 vs AoA



(b) Mass flow vs AoA



(c) Compressor map shift



(d) Stall margin degradation

Fig. 1 Aerothermodynamic degradation chain.

V. The 2025 Dubai Airshow Incident

A. Failure Chain Synthesis

1. Interpretation of Aerothermodynamic Degradation Chain

The coupled behavior of inlet aerodynamics and compressor performance is illustrated in Fig. 1. As shown in Fig. 1(a), the circumferential distortion coefficient (DC60) increases nonlinearly with angle of attack, reflecting the onset of vortex-dominated and separated flow at the inlet plane. This distortion produces a non-uniform total pressure field at the aerodynamic interface plane (AIP).

The resulting reduction in effective inlet area and flow quality leads to a measurable decrease in engine mass flow, as shown in Fig. 1(b). This reduction is not purely geometric, but arises from distortion-induced blockage and boundary layer ingestion, which together limit the usable flow delivered to the compressor.

The combined effects of pressure loss and reduced mass flow shift the compressor operating point toward the surge boundary, as illustrated in Fig. 1(c). Under these conditions, the compressor operates with reduced stability margin and

increased sensitivity to localized flow disturbances.

As shown in Fig. 1(d), the cumulative effect of distortion, pressure recovery loss, and thermal loading produces a progressive reduction in stall margin, with the system entering the failure regime beyond approximately 20° angle of attack. This indicates that under sustained maneuvering conditions, the compressor may exceed its stability limits, resulting in surge or stall. The behavior observed in Fig. 1 is consistent with the assumed blockage trend in Table 4.

2. Model Assumptions and Limitations

The results presented in Fig. 1 are derived from reduced-order models and representative parameter values based on established inlet and compressor performance relationships. These models are intended to capture first-order system behavior rather than exact engine performance under flight conditions. The distortion growth, effective inlet area reduction, and stall margin degradation are approximated using simplified functional relationships, and do not account for detailed three-dimensional flow structures, unsteady effects, or full engine control system response. Furthermore, the absence of publicly available flight-test data for the Tejas Mk1 inlet and F404-GE-IN20 compressor under high-angle-of-attack conditions necessitates the use of generalized compressor and inlet performance trends derived from literature. These results represent upper-bound system behavior under severe off-design conditions and are not intended as exact flight-test values.

Table 4 and Fig. 1 cover positive angles of attack only. The terminal negative-G segment (negative AoA, where the wing no longer shields the inlet lips) lies outside the range over which $\delta(\alpha)$ is defined. The model therefore quantifies stall margin erosion over the display sequence; the distortion level during the final push is assumed, not computed, to be at least comparable to the high positive-AoA case. Using representative inputs (SM_0 = 15–20%, η_r = 0.94, $DC60$ = 0.15, $T_0 \approx$ 309 K), the stall margin model yields a reduced but positive steady-state margin of approximately 6–11%. Loss of stability therefore requires a transient distortion component of at least this magnitude, consistent with instantaneous distortion peaks exceeding time-averaged DC60 during rapid maneuver transitions [6, 13].

B. Incident Overview

On November 21, 2025, a HAL Tejas Mk1 of the Indian Air Force was lost during a scheduled aerobatic display at the Dubai Airshow, resulting in the fatal injury of the pilot, Wing Commander Namansh Syal [23, 26, 27].

The aircraft took off at approximately 14:00 hrs local time and executed a sequence of standard demonstration maneuvers, including loops, rolls, and high-speed passes between approximately 14:05 and 14:09 [25, 33].

During the final phase of the display, at approximately 14:10 hrs, the aircraft entered a low-altitude maneuver involving a negative-G turn [29, 33]. Video and eyewitness accounts indicate that the aircraft failed to recover from this maneuver, rapidly losing altitude and entering a steep nose-down descent [24, 26]. The aircraft was observed attempting to regain level flight shortly before impact, but remained at an altitude of less than approximately 500 ft, providing

insufficient recovery margin [33]. The aircraft subsequently impacted the ground within seconds, resulting in a fatal accident [23, 26].

The maneuver sequence preceding the crash consisted of multiple high-energy and high-angle-of-attack segments executed in rapid succession, with limited steady-flight intervals between maneuvers. These conditions are characteristic of airshow display profiles, which impose sustained aerodynamic and thermal loading on the propulsion system, and form the basis for the failure mechanism analysis developed in the following section.

C. Atmospheric Conditions

On November 21, 2025, Dubai experienced ambient temperatures of approximately 29.4°C (85°F) with relative humidity averaging 46% for that period [21, 22]. The corrected air density under these conditions, computed from the ideal gas law with humidity correction at sea-level pressure, is:

$$\rho_{corr} \approx 1.158 \text{ kg/m}^3 \tag{14}$$

compared to ISA sea-level standard of 1.225 kg/m^3, representing approximately a 5.5% density reduction. This reduction directly decreases available mass flow through the engine and shifts compressor operating conditions away from design-point performance [1, 2].

D. Flight Timeline Reconstruction

Video analysis of the display sequence indicates that the aircraft initiated maneuvering approximately 15–20 s after takeoff. The initial maneuver involved a transition from a shallow climb angle of approximately 5–15° to a steeper climb of approximately 45°, executed over a period of approximately 2–3 s. During this climb phase, the aircraft performed two consecutive barrel roll maneuvers while maintaining positive climb, followed by continued climb at elevated angle of attack. Subsequent maneuvering included multiple roll sequences along the runway axis, where the aircraft momentarily held intermediate bank angles (approximately 90° and 180°) for durations on the order of 2–3 s, with rapid transitions between these positions.

After brief steady-flight intervals of approximately 1–2 s, additional roll maneuvers were performed, indicating rapid maneuver chaining with limited recovery time between successive high-load conditions. At a later stage, the aircraft executed a full vertical loop. Visible condensation over the wing surfaces during this maneuver indicates locally high lift coefficients and elevated aerodynamic loading. Following the loop, the aircraft descended at a steep flight path angle estimated to be approximately 60–80° before recovering toward level flight over approximately 5–10 s.

The aircraft then maintained near-level flight briefly before initiating further maneuvering, including a quarter roll followed by a sustained high-load maneuver lasting approximately 7–9 s. Additional roll maneuvers and sustained bank

angles in the range of approximately 45–90° were observed, with rapid transitions between attitudes. In a later segment, the aircraft maintained a high bank angle of approximately 80–90° while climbing at an estimated 35–50° flight path angle for a duration exceeding several seconds. This was followed by a continuous pull-through maneuver transitioning from climb to descent (approximately 25–45°), before returning to near-level flight within approximately 2–4 s.

Following a brief steady segment, the aircraft entered a maneuver involving a rapid pitch-up beyond vertical, followed by correction to approximately 90° attitude. The aircraft then executed a sustained negative-G maneuver lasting approximately 6–8 s, transitioning to an inverted orientation (approximately 180° bank). Negative-G maneuvers are known to impose significant aerodynamic and propulsion system challenges [29, 30].

Shortly after this maneuver, a loss of control was observed. The aircraft entered a nose-down descent at an estimated 20–40° flight path angle while inverted and at low altitude. Reports indicate the aircraft was operating below approximately 500 ft during the final phase, providing minimal recovery margin [25, 31]. Video evidence indicates an attempt to recover through roll correction toward upright orientation combined with a pull-up maneuver. However, the available altitude was insufficient to complete recovery, and the aircraft impacted the ground shortly thereafter [26].

Analysis of the display sequence implies that the maximum observed steady-flight interval between successive high-load maneuvers was approximately 10 s, with a conservative upper bound of 15 s applied in the present analysis to account for observational uncertainty. Given a characteristic turbine thermal time constant of 30–45 s for lightweight afterburning turbofan engines of the F404 class [9, 14], a 15 s recovery interval provides at most 28–39% thermal dissipation between maneuvers, assuming first-order decay ($1 - e^{-t/\tau}$). Successive high-load segments therefore occur under progressively elevated baseline thermal conditions, with a minimum of 61–72% of accumulated thermal load carried forward into each subsequent maneuver. This is consistent with the cumulative accumulation model presented in Fig. 2.

1. Inlet Distortion and Flow Degradation

As established in the preceding sections, the aircraft undergoes repeated high-angle-of-attack and high-bank-angle maneuvers. These conditions are known to produce significant inlet flow distortion, characterized by non-uniform total pressure distribution at the aerodynamic interface plane (AIP). Using representative distortion growth trends from literature, the distortion coefficient (DC60) is estimated to increase from nominal values of approximately 0.02 under clean conditions to values exceeding 0.10–0.15 under combined high-angle-of-attack and high-sideslip conditions [6, 7]. Such distortion levels are sufficient to induce localized compressor flow separation and reduce overall compressor stability margin [11].

2. Mass Flow Reduction

The effective mass flow rate is given by:

$$\dot{m}_{eff} = \rho A(1 - \delta)V \tag{15}$$

Using previously established values of $\rho = 1.158$ kg/m^3, $A = 0.50$ m^2, and $V \approx 114$ m/s, at high-angle-of-attack conditions with $\delta \approx 0.30$–0.36, the effective mass flow reduces to:

$$\dot{m}_{eff} \approx 1.158 \times 0.50 \times (1 - 0.35) \times 114 \approx 42.9 \text{ kg/s}$$

This represents an equivalent flow-capacity reduction of approximately 30–36% relative to the nominal engine mass flow of ~66 kg/s for the F404-GE-IN20 [17], sufficient to impose significant compressor operating point displacement toward the surge line, as illustrated in Fig. 1(c).

3. Compressor Operating Point Shift

The combined effects of reduced mass flow and increased inlet distortion result in a displacement of the compressor operating point toward the surge boundary. Distortion levels above the DC60 ≈ 0.06 onset threshold identified in Section III.A.2 reduce compressor stability margin, and values exceeding 0.10 reduce it substantially [8, 11]. Under these conditions, the compressor becomes increasingly sensitive to transient flow disturbances and non-uniform inlet conditions, particularly during rapid maneuver transitions.

4. Thermal Accumulation

During the observed maneuver sequence, high-load segments lasting approximately 5–10 s are interspersed with recovery intervals of only 1–4 s. Given typical turbine thermal time constants on the order of tens of seconds, these intervals are insufficient for complete thermal recovery [14]. As a result, the engine experiences cumulative thermal buildup:

$$T_{n+1} > T_n \tag{16}$$

This progressive increase in hot-section temperature is expected to reduce stall margin through transient rotor–casing thermal mismatch and increased compressor tip clearance under off-design operating conditions [10, 14]. This mechanism is not included in the reduced-order stall margin model and is presented here as a qualitative contributor; quantifying it requires engine-specific clearance data. The thermal response of the engine under sustained maneuvering is illustrated in Fig. 2.

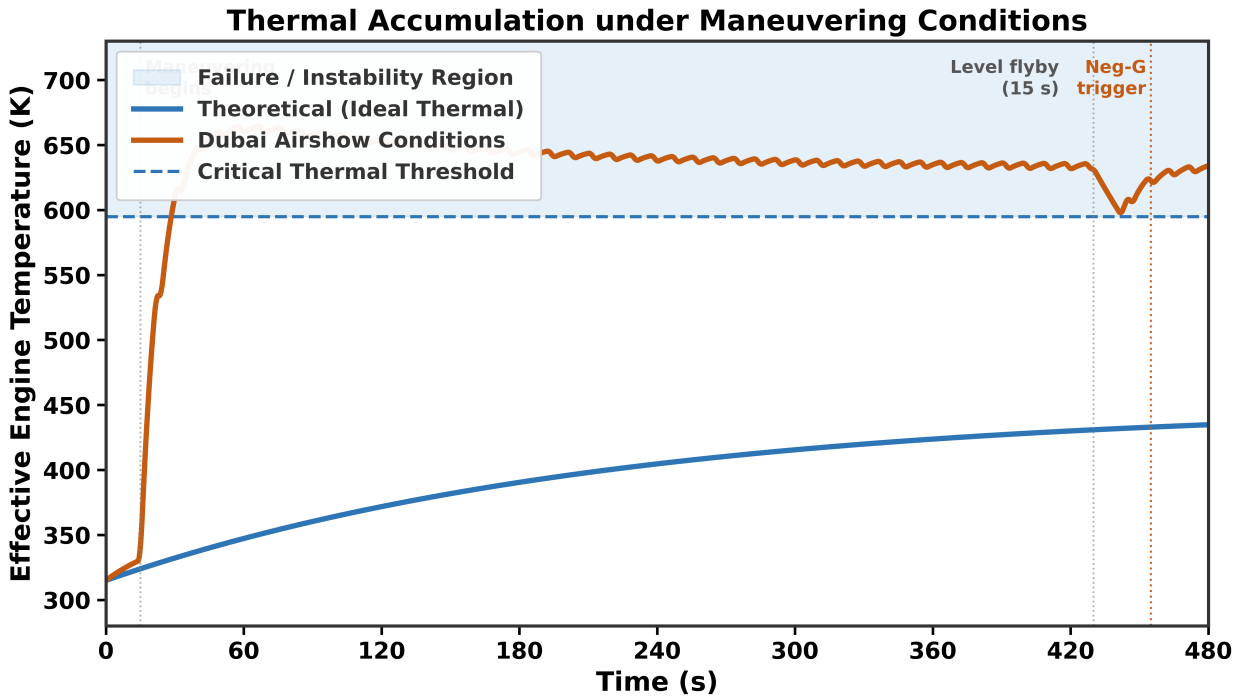


Fig. 2 Thermal accumulation under sustained maneuvering conditions.

5. Stall Margin Collapse and Failure

The combined effects of inlet distortion, reduced mass flow, and thermal accumulation result in a progressive reduction in stall margin, as shown in Fig. 1(d). A stall margin approaching approximately 10% represents the lower bound of stable compressor operation under nominal conditions. However, in the presence of significant distortion and transient loading, effective stall may occur at higher nominal margins [11].

Under the conditions described in this section, the propulsion system is driven into an unstable operating regime. The compressor is subjected to non-uniform flow, reduced mass flow, and elevated thermal loading, ultimately leading to compressor stall or surge. The resulting loss of thrust at low altitude provides insufficient recovery margin, leading to the observed loss of control and impact. This indicates that the final maneuver acted as a threshold trigger event within an already degraded propulsion system, rather than an isolated cause [13, 14].

VI. Proposed Redesign Alternatives

All proposed inlet configurations are evaluated with reference to the baseline HAL Tejas Mk1 airframe geometry and propulsion system. The engine mass flow requirements, flight conditions, and external aerodynamic environment are held constant across all configurations, isolating the effect of inlet geometry on distortion, mass flow, and compressor stability.

A. Diverterless Supersonic Inlet (DSI)

A Diverterless Supersonic Inlet (DSI) configuration is proposed as an intermediate redesign solution that combines improved aerodynamic performance with moderate implementation complexity.

1. Geometry and Flow Characteristics

The DSI configuration replaces the conventional splitter plate and diverter system with a three-dimensional compression bump integrated into the fuselage surface. This bump generates a controlled pressure gradient that passively displaces the boundary layer away from the inlet, eliminating the need for a physical diverter. The absence of a splitter plate reduces flow separation and simplifies the inlet geometry, while the compression surface improves flow conditioning prior to entry into the inlet duct.

2. Aerodynamic Impact

The aerodynamic benefits of the DSI configuration include passive boundary layer control without mechanical diverters, reduced total pressure loss due to simplified geometry, and lower distortion compared to conventional lateral inlets. Based on established DSI performance characteristics [4, 6], the distortion coefficient is expected to reduce to approximately $DC60 \approx 0.05$–0.08 under high-angle-of-attack conditions, with correspondingly higher retained mass flow relative to the baseline configuration.

3. Advantages and Limitations

Advantages include reduced mechanical complexity, improved aerodynamic efficiency, and lower distortion relative to the baseline. Limitations include the requirement for precise geometric optimization, performance sensitivity to design-point conditions, and moderate airframe redesign.

4. Projected Performance: DSI Configuration

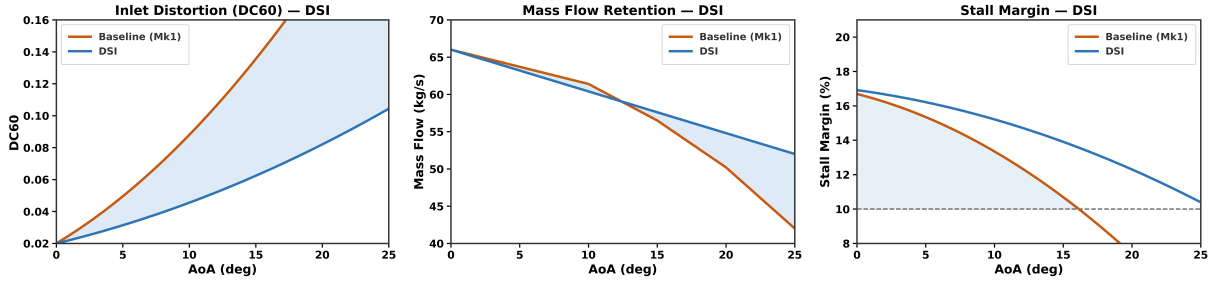


Fig. 3 Projected aerodynamic performance under Dubai maneuver conditions for a DSI configuration.

The DSI configuration demonstrates consistent improvement across all stages of the failure chain identified in Section V (Figs. 3 and 4). Distortion growth is significantly reduced with increasing angle of attack, resulting in

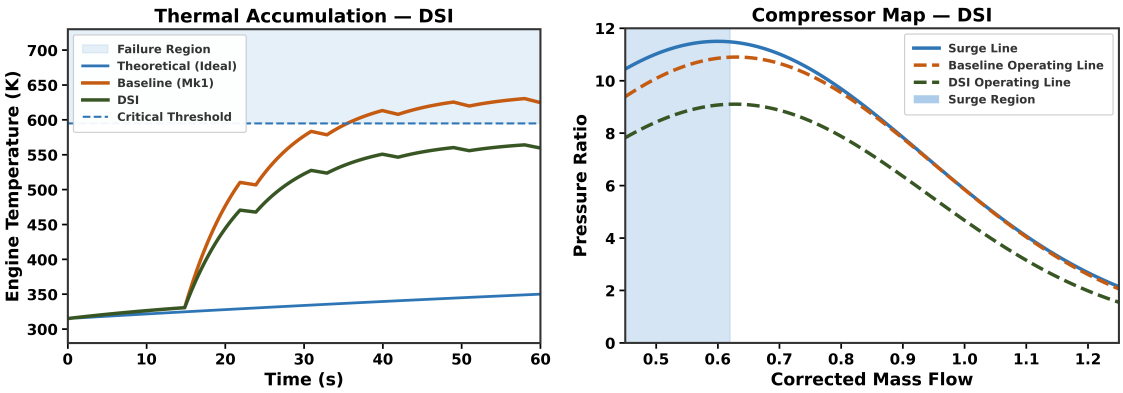


Fig. 4 Projected thermodynamic and compressor behavior for a DSI configuration.

improved total pressure uniformity at the AIP. The improved inlet flow quality leads to higher effective mass flow retention, which in turn delays thermal accumulation and prevents rapid degradation of compressor stability. As shown in Fig. 4, the compressor operating line remains further from the surge boundary compared to the baseline configuration, indicating improved robustness under off-design maneuvering conditions.

B. Chin / Ventral Intake

1. Geometry and Flow Characteristics

In this configuration, the inlet is positioned along the lower fuselage centerline, forward of the wing root. This placement isolates the inlet from vortex-dominated flow structures generated by the wing and leading edge extensions, resulting in a comparatively uniform incoming flow field. The absence of bifurcation eliminates S-duct-induced secondary flow and swirl distortion, reducing total pressure non-uniformity at the AIP.

2. Aerodynamic Impact

Based on representative inlet performance trends [4, 6], the distortion coefficient is expected to reduce to approximately $DC60 \approx 0.04$–0.06 under high-angle-of-attack conditions, compared to 0.12–0.15 for the baseline configuration, with correspondingly improved total pressure recovery and reduced sensitivity to angle-of-attack and sideslip.

3. Advantages and Limitations

Advantages include superior flow uniformity, reduced distortion-induced instability, and improved compressor operability. Limitations include increased radar cross-section, significant airframe redesign requirements, and integration constraints with landing gear and structure.

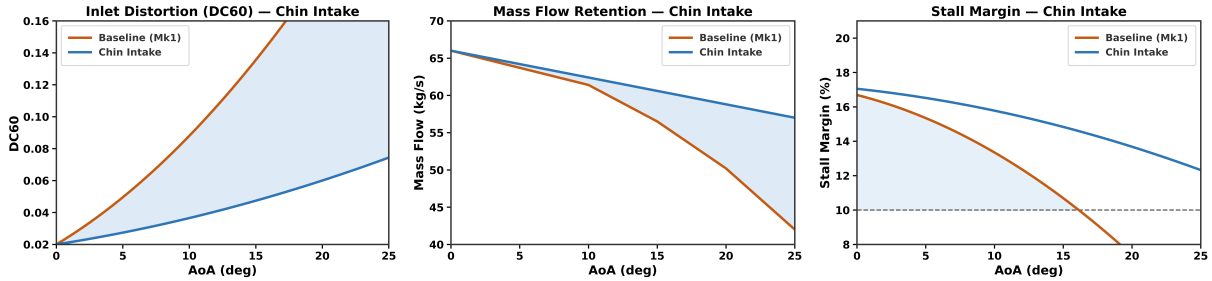


Fig. 5 Chin intake projected performance.

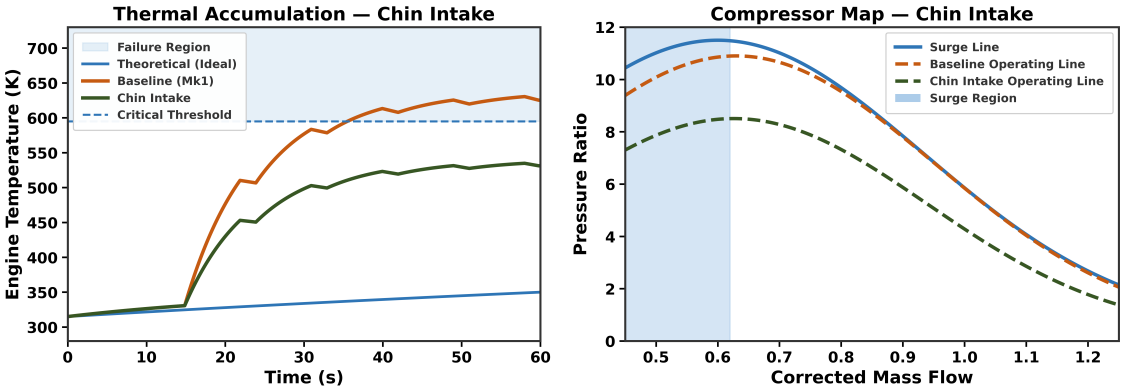


Fig. 6 Projected thermodynamic and compressor behavior for a chin intake configuration.

4. Projected Performance: Chin Intake Configuration

The chin intake configuration (Figs. 5 and 6) provides the most favorable inlet flow conditions among the configurations considered, resulting in minimal distortion, improved mass flow retention, and the highest available stall margin. These effects collectively maintain the compressor operating point well within the stable regime, even under aggressive maneuvering conditions.

C. Rear-Shifted Lateral Intake

A modified lateral inlet configuration is proposed in which the inlet location is shifted aft of the wing root to reduce exposure to vortex-dominated flow regions.

1. Geometry and Flow Characteristics

In the baseline configuration, the inlet is positioned at the wing root, where strong vortex structures and flow separation are prevalent at high angles of attack. By shifting the inlet aft along the fuselage, the influence of these vortical structures is reduced. However, the boundary layer thickness increases along the fuselage, resulting in a trade-off between reduced vortex ingestion and increased low-energy boundary layer ingestion.

2. Aerodynamic Impact

Representative estimates suggest a reduction in distortion levels to approximately $DC60 \approx 0.08$–0.10, with corresponding reductions in effective blockage factor compared to the baseline configuration. Moderate reduction in circumferential distortion and partial improvement in pressure recovery are expected, accompanied by increased sensitivity to boundary layer ingestion.

3. Advantages and Limitations

Advantages include minimal structural redesign, lower implementation cost, and reduced vortex interaction. Limitations include residual distortion under off-design conditions, increased boundary layer ingestion, and limited improvement in extreme maneuvering regimes.

4. Projected Performance: Rear-Shifted Lateral Intake

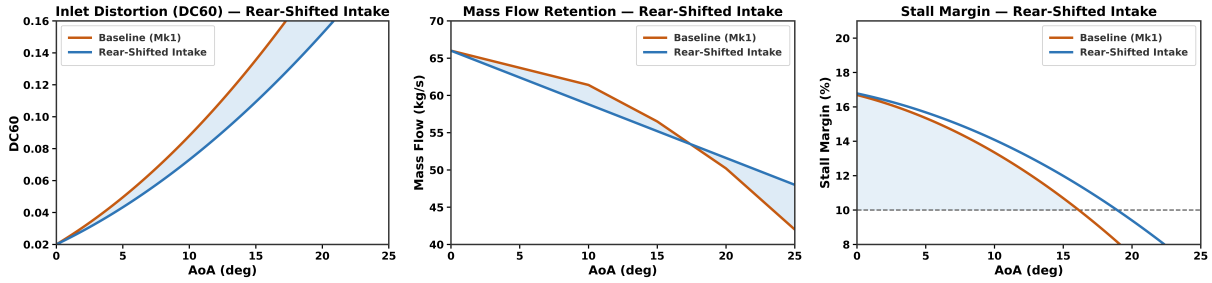


Fig. 7 Rear-shifted intake projected performance.

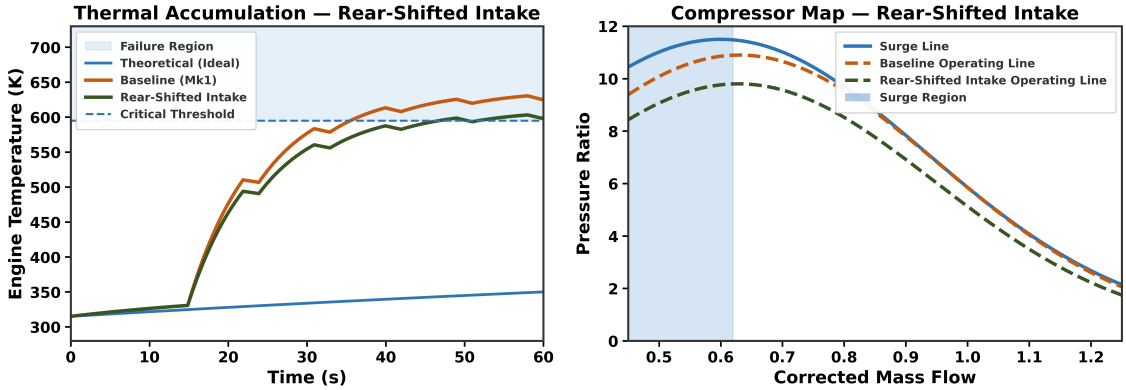


Fig. 8 Projected thermodynamic and compressor behavior for a rear-shifted intake.

The rear-shifted configuration (Figs. 7 and 8) demonstrates moderate improvement across the failure chain. Distortion growth is reduced due to decreased interaction with wing-root vortex structures. However, the increased boundary layer thickness introduces low-energy flow into the inlet, limiting the overall improvement in mass flow retention. As a result, thermal accumulation is reduced but not eliminated, and the compressor operating point remains closer to the surge

boundary than in the DSI and chin intake configurations. The following limitations apply to this analysis: boundary layer growth is approximated using simplified trends; the interaction between residual vortex structures and the shifted inlet position is not fully resolved; and structural and aerodynamic integration constraints associated with relocating the inlet are not modeled. Despite these limitations, the results provide a consistent comparative assessment relative to the baseline and alternative designs.

D. Comparative Performance Analysis

Table 5 Comparative Inlet Performance Under Dubai Maneuver Conditions

Configuration	DC60 Range	Mass Flow (kg/s)	Mass Flow Loss	Stall Margin (%)	Thermal Behavior
Baseline (Mk1)	0.12–0.15	42–46	30–36%	9–11	Rapid accumulation, exceeds threshold
Rear-Shifted Lateral	0.08–0.11	48–54	18–27%	11–13	Reduced accumulation, still critical
DSI Configuration	0.05–0.08	52–58	12–21%	12–14	Delayed accumulation, near-safe
Chin Intake	0.04–0.06	56–62	6–15%	14–16	Minimal accumulation, stable

The comparative results in Table 5 are derived from the reconstructed Dubai maneuver conditions and corresponding reduced-order inlet and propulsion models. The values are consistent with established trends in intake distortion, compressor response, and thermal behavior reported in the literature [3, 4, 6]. The analysis demonstrates that inlet geometry has a first-order influence on propulsion system stability under off-design conditions. While incremental modifications such as inlet relocation provide partial improvement, more fundamental geometric redesigns, including DSI and chin intake configurations, offer significantly greater robustness against distortion-induced compressor instability. These trends indicate that the failure mechanism identified in Section V is sensitive to inlet geometry and can be mitigated through appropriate aerodynamic design modifications [4, 13].

VII. Conclusion

This study investigated the propulsion system behaviour of the HAL Tejas Mk1 under the reconstructed maneuver conditions observed during the Dubai Airshow incident of November 21, 2025. A combined aerodynamic and thermodynamic analysis was performed to evaluate the interaction between inlet flow characteristics and compressor stability [2, 4, 6].

The results demonstrate that the propulsion system is highly sensitive to inlet distortion, reduced mass flow,

and cumulative thermal loading under sustained high-angle-of-attack maneuvering. The combined effect of these mechanisms results in a progressive reduction in compressor stall margin, consistent with a trajectory toward an unstable operating condition [6, 13, 14].

Comparative analysis of alternative inlet configurations shows that inlet geometry has a first-order influence on propulsion system stability [4]. Incremental modifications, such as rearward relocation of the lateral inlet, provide moderate improvement but do not fully eliminate the identified instability mechanisms. In contrast, geometrically distinct configurations — including Diverterless Supersonic Inlets (DSI) and chin intake designs — demonstrate substantially improved flow uniformity, higher mass flow retention, and increased stall margin under identical operating conditions [1, 6].

These findings indicate that the observed failure mechanism is not an intrinsic limitation of turbofan propulsion systems, but rather a result of inlet–flow interaction under off-design conditions. While definitive causation requires access to flight-test and telemetry data held by the Court of Inquiry [23], the present analysis establishes a physically consistent pathway to compressor instability under the documented ambient, aerodynamic, and thermal conditions [12, 13, 21].

The study highlights the importance of inlet design in maintaining propulsion system robustness during aggressive maneuvering and provides a framework for evaluating inlet–engine interaction in future aircraft configurations [3, 4]. Future work should incorporate high-fidelity computational fluid dynamics and experimental validation to further quantify inlet–compressor interaction under transient maneuvering conditions, and the findings of the Court of Inquiry should be considered alongside this analysis when conclusions are drawn about the Dubai incident.

The analysis presented is based on publicly available data and reconstructed flight conditions, and does not represent an official accident determination. The analysis does not assert causation, but identifies a consistent mechanism through which inlet–engine interaction may reduce propulsion stability under the conditions documented at Dubai on November 21, 2025.

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Acknowledgments

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This paper is dedicated to the memory of Wing Commander Namansh Syal, Indian Air Force, who gave his life

representing India at Dubai on November 21, 2025 [26]. He was a graduate of Sainik School Sujanpur Tira and of the National Defence Academy (115th Course), where he was awarded the President's Silver Medal and led the Passing-Out Parade of November 2008 [36, 37]. He was commissioned into the IAF in December 2009 and flew MiG-21s, Su-30 MKIs, and the Tejas over a sixteen-year career [36]. He was one of the few IAF pilots qualified to demonstrate the Tejas at an international airshow [36]. His father, Nb Sub Jagan Nath Syal (Retd), said: *"He never disappointed me. He never had a dull moment in his life and won every competition that he participated in"* [39].

The analysis in this paper exists because he deserved a technically rigorous account of what the physics says may have happened, in the public record, where it cannot be set aside.

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